LEAF-MARKED FIRST PASSAGE ON UNEQUAL FINITE SPIDERS: A RATIONAL TRANSFORM, SHARP EXTREMIZERS, AND A COARSE-DATA INVERSE

ANONYMOUS

Abstract. Join $r \geq 2$ finite paths of positive integer lengths at one centre, start simple random walk at the centre, and absorb it at the first leaf. We derive the complete probability generating function jointly marked by the absorbing leaf for arbitrary unequal arms. A centre-excursion renewal reduces the law to one-dimensional continuants and gives a common rational denominator for all leaf marks. Besides parity and the first possible atom, the transform yields the compact formula

$$\mathrm{Var}(T) = \frac{C-2L}{3H} + \frac{L^2}{3H^2}, \qquad H = \sum_i \ell_i^{-1}, \quad L = \sum_i \ell_i, \quad C = \sum_i \ell_i^3.$$

For fixed arm count and total length, strict integer transfers identify the exact mean-minimizing profile—one long arm and all other arms of length one—and the exact mean-maximizing balanced profiles. The labelled endpoint law determines precisely the primitive arm-length ratios and is blind to a common dilation; adding the mean recovers that scale uniquely. General-tree endpoint and mean formulas, an unequal-arm endpoint exercise, equal-arm star laws, spider spectral methods, and general inverse first-passage frameworks are treated as prior background. An exact-arithmetic audit performs 1,446,432 independent recurrence, coefficient, moment, extremal, and inverse checks.

1. Process, ownership boundary, and theorem

For positive integers $\ell_1, \ldots, \ell_r$, let $\mathcal{S}(\ell_1, \ldots, \ell_r)$ be the tree formed by identifying one endpoint of r paths of edge lengths ℓ_i. Call the identified vertex the *centre*, label the other endpoints $1, \ldots, r$, and make those leaves absorbing. Simple random walk starts at the centre. Write T for its first leaf-hitting time and I for the label of the leaf it hits.

Unequal arms obstruct the radial lumping available when all lengths agree: the time and the absorbing label remain coupled. The paper resolves that coupling through a leaf-marked excursion transform. The contribution is a four-part conjunction: the arbitrary unequal-arm marked transform, its compact variance, sharp fixed-total extremizers, and a geometry inverse from deliberately coarse data. Table 1 states the corresponding subtraction.

Table 1. Primary-source subtraction. The middle column receives zero contribution credit; the right column is the narrower residual studied here.

Source	Owned input	Boundary retained here
Pearce [5]	finite trees with absorbing leaves; endpoint probabilities and expected walk length	neither endpoint mass nor mean is claimed
Pal–Mesikepp, Problem 2.4 [4]	designated-leaf probability for a star with unequal integer arms	endpoint law is used only for extrema and inversion
Castella–Sericola [1]	hitting distributions and moments for equal-length finite stars	the transform axis requires arbitrary unequal arms and a leaf mark
de la Iglesia–Juarez [2]	spectral and stochastic factorization methods for half-line spider chains	no spectral-framework or generic-resolvent claim
de la Peña–Gzyl–McDonald [3]	unknown transitions on a known augmented tree recovered from rich boundary time/place data	fixed simple-walk kernel and unknown integer arms, using only endpoint probabilities plus one mean

The term "reflecting–absorbing factorization" in [2] denotes a stochastic block factorization and a Darboux transformation, not absorption at finite leaves. Conversely, the inverse theorem in [3] observes complete joint time/place laws on two boundary layers and recovers transition probabilities; our topology class and transition rule are fixed, while the integer arm lengths are unknown. These distinctions prevent generic spider, resolvent, and tomography language from carrying contribution credit.

Define continuant polynomials by

$$P_0(z) = 0, \qquad P_1(z) = 1, \qquad P_2(z) = 2, \qquad P_m(z) = 2P_{m-1}(z) - z^2 P_{m-2}(z). \tag{1}$$

Put

$$P(z) = \prod_{j=1}^{r} P_{\ell_j}(z), \qquad D(z) = rP(z) - z^2 \sum_{i=1}^{r} P_{\ell_i - 1}(z) \prod_{j \neq i} P_{\ell_j}(z). \tag{2}$$

Also write

$$H = \sum_{i=1}^{r} \frac{1}{\ell_i}, \qquad L = \sum_{i=1}^{r} \ell_i, \qquad C = \sum_{i=1}^{r} \ell_i^3. \tag{3}$$

Theorem 1 (marked law, moments, extrema, and inverse)**.** *For the walk on $\mathcal{S}(\ell_1, \dots, \ell_r)$, the following statements hold.*

(1) *For every labelled leaf i,*

$$F_i(z) := \mathbb{E}\big[z^T \mathbf{1}_{\{I=i\}}\big] = \frac{z^{\ell_i} \prod_{j \neq i} P_{\ell_j}(z)}{D(z)}. \tag{4}$$

Its support lies in $\ell_i + 2\mathbb{Z}_{\geq 0}$, and

$$\mathbb{P}(T = \ell_i, I = i) = \frac{1}{r 2^{\ell_i - 1}}. \tag{5}$$

(2) *The endpoint and mean identities*

$$\mathbb{P}(I = i) = \frac{\ell_i^{-1}}{H}, qquad \mathbb{E}T = \frac{L}{H} \tag{6}$$

are background inputs from the general-tree and unequal-arm literature in Table 1. The residual second-moment conclusion is

$$\mathrm{Var}(T) = \frac{C - 2L}{3H} + \frac{L^2}{3H^2}. \tag{7}$$

(3) *Fix r and $L \geq r$, and write $L = qr + s$ with $0 \leq s < r$. Then*

$$\frac{L}{r - 1 + 1/(L - r + 1)} \leq \mathbb{E}T \leq \frac{L}{(r - s)/q + s/(q + 1)}. \tag{8}$$

Equality on the left holds exactly for permutations of $(L - r + 1, 1, \dots, 1)$; equality on the right holds exactly when the arms are balanced, with $r - s$ lengths q and s lengths $q + 1$.

(4) *The labelled endpoint vector determines the primitive positive integer arm vector $d = (d_i)$, while $\ell = cd$ is invisible for every common positive integer dilation c. Adding $\mathbb{E}T$ determines the unique scale by*

$$c^2 = \mathbb{E}T \, \frac{\sum_i d_i^{-1}}{\sum_i d_i}. \tag{9}$$

The theorem assumes from the outset that the unknown object is a labelled finite spider and that its transition kernel is simple random walk. Part (4) does not recover an arbitrary tree or unknown transition probabilities.

2. Killed paths and centre renewal

Let U_m denote the Chebyshev polynomial of the second kind, with $U_{-1} = 0$. Recurrence (1) is equivalently the polynomial identity

$$P_m(z) = z^{m-1} U_{m-1}(1/z). \tag{10}$$

The Chebyshev notation is a convenient solution of a second-order recurrence; neither generic continuants nor gambler's ruin are contribution claims.

Lemma 2 (one killed arm)**.** *On $\{0,1,\dots,m\}$, start fair nearest-neighbour walk at 1 and stop on first hitting $\{0,m\}$. If τ is this path time, then*

$$\mathbb{E}_1[z^\tau \mathbf{1}_{\{X_\tau=m\}}] = \frac{1}{U_{m-1}(1/z)} = \frac{z^{m-1}}{P_m(z)}, \tag{11}$$

$$\mathbb{E}_1[z^\tau \mathbf{1}_{\{X_\tau=0\}}] = \frac{U_{m-2}(1/z)}{U_{m-1}(1/z)} = \frac{zP_{m-1}(z)}{P_m(z)}. \tag{12}$$

Proof. For $1 \le k \le m-1$, either boundary-marked transform u_k satisfies

$$u_k = \frac{z}{2}(u_{k-1} + u_{k+1}).$$

For absorption at m, the boundary values are $(u_0, u_m) = (0,1)$, and the solution is $u_k = U_{k-1}(1/z)/U_{m-1}(1/z)$. For absorption at zero, the boundary values are $(1,0)$, and the solution is $u_k = U_{m-k-1}(1/z)/U_{m-1}(1/z)$. Evaluating at $k=1$ and applying (10) proves both identities. For $m=1$, the formulas use $P_0 = 0$ and express immediate success after the later-added centre step. □

Proof of Theorem 1(1). An attempt from the centre chooses arm j, makes one step to its first vertex, and then either reaches leaf j or returns to the centre. By Lemma 2, the transform of a successful attempt at leaf i is

$$S_i(z) = \frac{1}{r}\frac{z^{\ell_i}}{P_{\ell_i}(z)},$$

and the transform of an unsuccessful attempt, with the arm unmarked, is

$$Q(z) = \frac{z^2}{r}\sum_{j=1}^{r}\frac{P_{\ell_j-1}(z)}{P_{\ell_j}(z)}.$$

After every failure the walk is again at the centre, so the strong Markov property gives the formal renewal series

$$F_i(z) = S_i(z)\sum_{k\ge 0} Q(z)^k = \frac{S_i(z)}{1-Q(z)}.$$

Multiplying numerator and denominator by $r\prod_j P_{\ell_j}(z)$ gives (4).

Each P_m is a polynomial in z^2. Equation (4) therefore has only powers congruent to ℓ_i modulo two, in agreement with bipartiteness. At the first possible time ℓ_i, the walk must choose arm i and then follow its unique shortest path without backtracking. The probability is $(1/r)(1/2)^{\ell_i-1}$, proving (5). □

3. Excursion moments and the variance

The renewal proof also separates the second moment into one-arm quantities. This route avoids treating differentiation of the common denominator as a black box.

Lemma 3 (one-attempt moments)**.** *Condition on choosing an arm of length m. Let A_m be the duration from departure at the centre until the attempt next reaches either the centre or the leaf, including the departure step, and let R_m be the event of return to the centre. Then*

$$\mathbb{P}(R_m^c) = \frac{1}{m}, \qquad \mathbb{E}A_m = m, \tag{13}$$

$$\mathbb{E}A_m^2 = \frac{m(m^2+2)}{3}, \qquad \mathbb{E}[A_m \mathbf{1}_{R_m}] = \frac{2(m^2-1)}{3m}. \tag{14}$$

Proof. Including the centre step, the success and return transforms from Lemma 2 are

$$A_m^{\mathrm{suc}}(z) = \frac{z^m}{P_m(z)}, \qquad A_m^{\mathrm{ret}}(z) = \frac{z^2 P_{m-1}(z)}{P_m(z)}.$$

Differentiating (1) and inducting gives

$$P_m(1) = m,$$

$$P_m'(1) = -\frac{m(m-1)(m-2)}{3},$$

$$P_m''(1) = \frac{m(m-1)(m-2)(m^2-7m+7)}{15}.$$

Substitution into the two quotient derivatives gives $A_m^{\mathrm{suc}}(1) = 1/m$, $(A_m^{\mathrm{ret}})(1) = 1 - 1/m$,

$$(A_m^{\mathrm{suc}} + A_m^{\mathrm{ret}})'(1) = m,$$

$$(A_m^{\mathrm{suc}} + A_m^{\mathrm{ret}})''(1) + (A_m^{\mathrm{suc}} + A_m^{\mathrm{ret}})'(1) = \frac{m(m^2+2)}{3},$$

and $(A_m^{\mathrm{ret}})'(1) = 2(m^2-1)/(3m)$. These are precisely (13)–(14). The identities remain valid at $m = 1$. □

Proof of Theorem 1(2). Average one attempt over the uniform choice of arm. Let B indicate a successful attempt and A its duration. Lemma 3 gives

$$p := \mathbb{E}B = \frac{H}{r}, \quad \mu := \mathbb{E}A = \frac{L}{r}, \quad \nu := \mathbb{E}A^2 = \frac{C+2L}{3r}, \quad \rho := \mathbb{E}[A(1-B)] = \frac{2(L-H)}{3r}. \tag{15}$$

Let T' be an independent copy of T, used after a failed attempt. Then $T = A + (1-B)T'$. First moments give $p\mathbb{E}T = \mu$, which recovers the prior mean in (6). Squaring before taking expectations gives

$$p\mathbb{E}T^2 = \nu + 2\rho\mathbb{E}T.$$

Using (15) and subtracting $(L/H)^2$ yields (7). Finally, evaluating (4) at $z = 1$ gives the prior endpoint identity in (6). □

4. Sharp fixed-mass extrema

The prior mean identity becomes a residual result when optimized with exact integer equality classes.

Proof of Theorem 1(3). Because L is fixed, minimizing $\mathbb{E}T = L/H$ is equivalent to maximizing $H = \sum_i 1/\ell_i$. If $2 \le a \le b$, the outward unit transfer $(a,b) \mapsto (a-1, b+1)$ changes the reciprocal sum by

$$\frac{1}{a(a-1)} - \frac{1}{b(b+1)} > 0.$$

Repeated outward transfers end only when all but one arm have length one. Strictness proves that the maximizers of H are exactly the permutations of $(L-r+1, 1, \dots, 1)$. Substitution gives the left side of (8).

For the other direction, if $b \ge a+2$, the inward transfer $(a,b) \mapsto (a+1, b-1)$ changes the old reciprocal sum minus the new one by

$$\frac{1}{a(a+1)} - \frac{1}{b(b-1)} > 0.$$

Thus H decreases strictly until all arm lengths differ by at most one. Writing $L = qr + s$ forces $r - s$ lengths q and s lengths $q+1$; substituting their reciprocal sum gives the right side of (8). These strict transfers also prove both equality classifications. If $L = r$, both classes reduce to the all-one profile, so the boundary case is included. □

5. What the coarse data identify

Proof of Theorem 1(4). Let $\pi_i = \mathbb{P}(I = i)$. The background endpoint identity gives

$$\frac{\pi_i}{\pi_j} = \frac{\ell_j}{\ell_i}. \tag{16}$$

Hence the labelled vector π determines the rational ray containing (ℓ_i) and therefore its unique primitive positive integer representative $d = (d_i)$. It cannot determine scale: every cd, $c \in \mathbb{Z}_{>0}$, has the same endpoint vector. Conversely, if two labelled spiders have the same endpoint vector, (16) makes their arm vectors proportional; primitive integer normalization shows that common dilation is the entire ambiguity.

For $\ell = cd$, the prior mean formula becomes

$$\mathbb{E}T = c^2 \frac{\sum_i d_i}{\sum_i d_i^{-1}}.$$

Solving gives (9). For data generated by an integer spider, the right-hand side is the square of its unique positive integer scale. No claim is made for arbitrary noisy vectors, unknown topology, or unknown transition kernels. □

6. Exact audit, limitations, and conclusion

A standalone standard-library verifier checks the formulas in exact integer and rational arithmetic. For every ordered profile with $2 \leq r \leq 5$ and $1 \leq \ell_i \leq 4$, it compares a literal vertex-state Markov recursion with the rational series (4) coefficient by coefficient through time $2L + 12$. Independent routes check quotient derivatives, the excursion moments, every positive composition with $2 \leq r \leq 6$ and $L \leq 24$ for the sharp bounds, primitive-ray recovery through arm length eight, and the equal-arm radial collapse as an owned control. Table 2 records the frozen run.

Table 2. Exact-arithmetic falsification audit. Counts are assertions, not proof steps or evidence of novelty.

Control	Assertions
Continuant recurrence and closed coefficients	400
Literal marked coefficients and mass	524,816
Endpoint/mean/variance derivatives	10,448
Independent one-attempt moments	400
Fixed-mass inequalities and equality classes	760,104
Primitive ray, dilation, and scale recovery	149,760
Equal-arm owned-background collapse	504
Total	1,446,432

The results have four deliberate limitations. First, the topology class is known and consists only of labelled spiders with positive integer arm lengths. Second, transitions are the unweighted simple-walk transitions; weighted or biased arms are outside the theorem. Third, the inverse uses exact population quantities and gives no estimator, stability bound, or noisy-data guarantee. Fourth, the paper does not address cover times, general trees, or equal-arm distribution theory. A bounded primary-source search did not locate the specific residual conjunction, but a search non-hit is not a novelty, priority, or freedom-to-operate certificate.

Within those boundaries, centre renewal gives a complete marked law rather than an unmarked recurrence, while the prior endpoint and mean identities support two additional exact questions: which integer geometries extremize the clock, and which geometry information survives coarse observation. The answers are respectively the strict unbalanced/balanced equality classes and the exact common-dilation boundary.

Declarations

Data availability. No external dataset is used. The paper-local exact verifier and its frozen text output accompany this anonymous internal manuscript.
Ethics statement. The work is mathematical and uses no human participants, personal data, animals, or interventions.
Author contributions. The anonymous author or authors performed the conceptualization, derivations, exact verification, source audit, and manuscript preparation. Identity and individual role allocation remain suppressed for anonymous review.
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